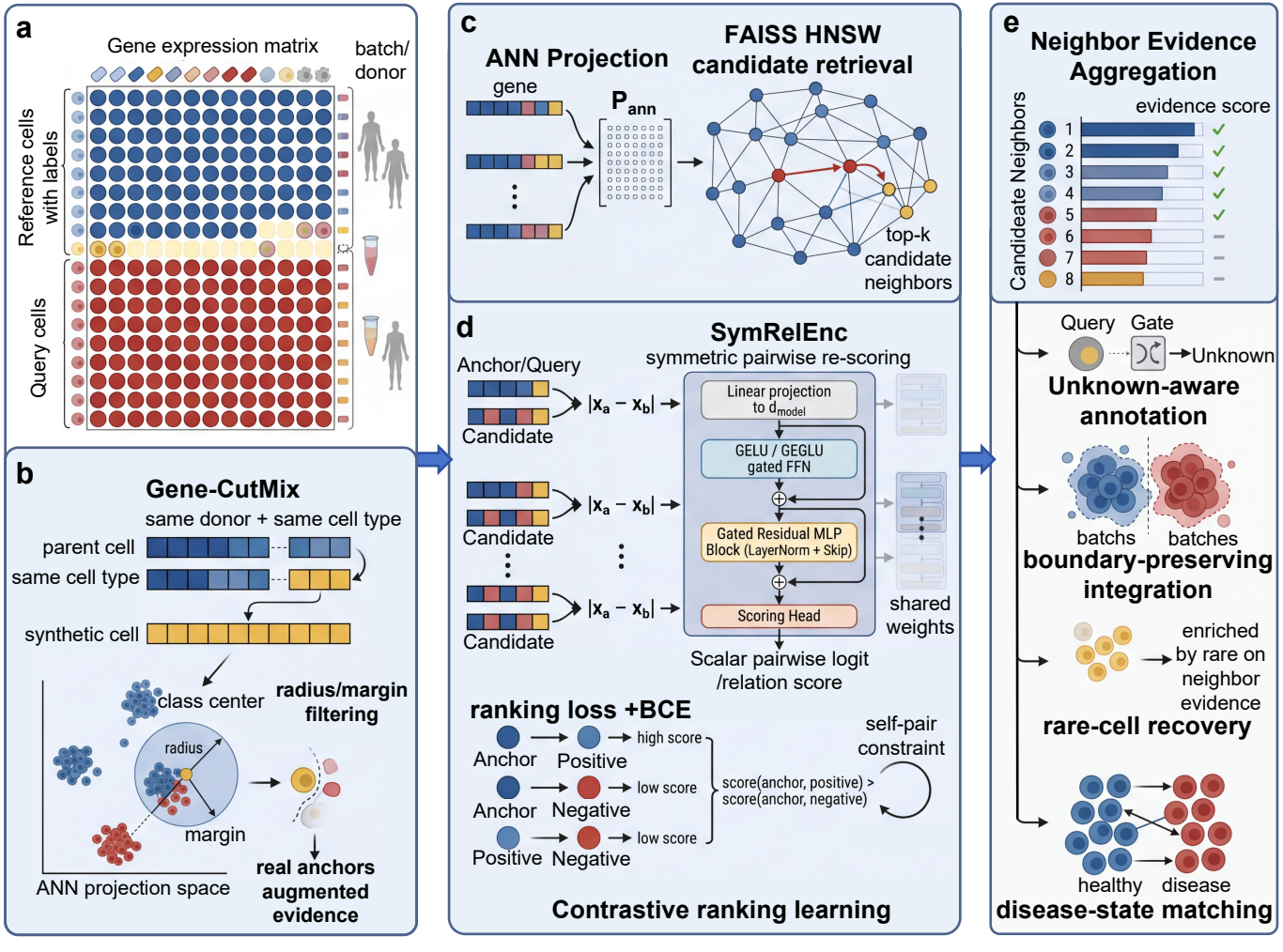

a
Gene expression matrix
batch/ donor
Reference cells with labels
Query cells
b
Gene-CutMix
same donor + same cell type
parent cell
same cell type
synthetic cell
class center
radius/margin filtering
radius
margin
ANN projection space
real anchors augmented evidence
c
ANN Projection
gene
P_{ann}
FAISS HNSW candidate retrieval
top-k candidate neighbors
d
SymRelEnc
symmetric pairwise re-scoring
Anchor/Query
Candidate
$|x_a - x_b|$
Linear projection to d_{model}
GELU / GEGLU gated FFN
Gated Residual MLP Block (LayerNorm + Skip)
Scoring Head
shared weights
Scalar pairwise logit /relation score
ranking loss +BCE
Anchor
Positive
Negative
high score
low score
score(anchor, positive) > score(anchor, negative)
self-pair constraint
Contrastive ranking learning
e
Neighbor Evidence Aggregation
Candideate Neighbors
evidence score
Query
Gate
Unknown
Unknown-aware annotation
batchs
batches
boundary-preserving integration
enriched by rare on neighbor evidence
rare-cell recovery
healthy
disease
disease-state matching